

Roan Kathrina Dimaculangan

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OBJECTIVE

Experienced Full-stack Developer with 8 years of experience in designing, implementing, testing, and maintaining web applications and a proven track record of excellence in creating innovative automations and leveraging expertise in multiple programming languages. Eager to contribute in a forward-thinking tech organization where I can leverage my expertise in full-stack development, along with my skills in machine learning, design thinking and data analytics to develop cutting-edge solutions and contribute to the growth of an innovative tech company.

HIGHLIGHT OF QUALIFICATIONS

Full-Stack Development:

- Actively engaged in the design and implementation of multiple applications covering both frontend (HTML, CSS, JavaScript (Vanilla, ReactJS, Angular) and backend (Python and Node.js) aspects, which were key deliverables across several project releases.
- Performed development and integration of public REST API endpoints to facilitate data retrieval of systems

Process Automation:

- Gained multiple recognitions for implementing process automation using Python and Linux Shell Scripting, leading to significant time savings and efficiency gains for the team.

Machine Learning and Data Analytics:

- Utilized Python and data analytics tools to implement machine learning models and automate data processing, enhancing decision-making and predictive analysis

Technical Leadership:

- Managed customer communications via email and video conferencing, ensuring clear and effective dialogue with stakeholders.
- Developed and maintained comprehensive Test Plan documents, ensuring project alignment with quality standards.
- Crafted detailed Project Analysis reports to inform decision-making and strategic planning.
- Allocated tasks and monitored the progress of project team members, ensuring timely and successful project delivery.

TECHNICAL SKILLS

Web Development:

Front-End:

Languages: **HTML, CSS, JavaScript (ES6+), PHP**

Frameworks/Libraries: **React JS, Angular (versions 5 and 8), Vanilla JS**

Tools: NPM, Developer Tools (FOR Responsive Web Design, Cross-Browser Compatibility and Performance Testing)

Back-End:

Languages: **Python (for scripting), Bash (for scripting), Node.js, Java**

Frameworks: **Django, Flask, FastAPI**

Databases: MongoDB, MySQL, REDIS

API Development: **RESTful API design and implementation**

DevOps and System Administration:

Version Control: **Git, Bitbucket**

CI/CD: TeamCity, Urban Deploy, Artifactory, **Docker**

Server Management: Linux server management and monitoring

Cloud Computing:

AWS Services: **Route 53, S3, EC2**

AWS: AWS infrastructure configuration

GCP: Test Environment Setup, VM Instances Management, Cloud Deployment

Data Science and Analytics:

Tools: **Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch**

Natural Language Processing (NLP): **Query Decomposition**, Evaluating Chatbot Responses

RAG (Retrieval-Augmented Generation) Technology, LLM Metrics (Sentiment Score, ROUGE, BLEU)

Automated Testing:

Selenium for Chatbot Agent Testing

Software Development Practices:

Methodologies: TDD, Waterfall, Agile/Scrum

EDUCATION

Postgraduate, Cloud Computing Technologies

September 2024 – Present

George Brown College | Toronto, ON, Canada

- Relevant Courses: Linux Shell Scripting, Windows Server Administration, Networking Fundamentals

Postgraduate, Applied A.I. Solutions Development

September 2023 – August 2024

George Brown College | Toronto, ON, Canada

- Official GPA: 3.70 out of 4.0
- Completed with Honors
- Relevant Courses: Machine and Deep Learning, Big Data

Bachelor of Science in Computer Engineering

April 2006 - June 2011

De La Salle University | Metro Manila, Philippines

- 1st Honor Dean's Lister 3rd term, SY 2009 - 2010
- 2nd Honor Dean's Lister 1st term, SY 2010 - 2011

PROFESSIONAL EXPERIENCE

I.T. Consultant (Part-time)

September 2022 – Present

Philippine Dealing System Holdings (PDS) and Philippine Dealing & Exchange Corp. (PDEX) | Quezon City, Philippines/ Remote

Project Summary: Provided strategic support in re-engineering PDS Legacy Systems, integrating international standards, and enhancing data processes for improved operational efficiency across financial applications.

- **System Code Analysis:** Conducted in-depth analysis of PDS legacy codebases, offering strategies for system modernization and developing a comprehensive code plan.
- **Production Support:** Addressed production issues across applications (HTML, CSS, JavaScript, JSP, Python) to ensure smooth daily operations of legacy systems, working closely with Application Support Management.
- **Bloomberg Integration:** Developed a monetized REST API within Bloomberg's IT infrastructure, enabling secure data retrieval (e.g., BVAL Rates, Corporate Board Summary) using Python.

- **PDS Billing System Overhaul:** Led data integration efforts, executing ETL processes to calculate and feed data into SAP, utilizing Python and Bash for optimized data handling.
- **ISIN Project:** Implemented updates across WordPress, Python, PHP, JSP, and SQL to align PDS security codes with ISIN standards, enhancing compliance with global securities protocols.
- **Data Format Transformation:** Converted BAP MART GS Repo trade files from CSV to XML as per Bloomberg standards, using Python and Django for backend processing.
- **Forward Bonds Project:** Managed ETL processes for bond data, transforming and loading security trade details from XML to REDIS, with end-of-day data integration into MySQL using Python and Django.

Technologies: Python, Django, Bash Scripting, RESTful APIs, WordPress, PHP, JSP, MySQL, REDIS, XML

Front-end Developer (Contract)

May 2024 – October 2024

Hank Zarihs (UK Financial Firm) via Kuubiik Ltd. | UK/ Remote

Project Summary: Created a centralized platform integrating UK financial APIs for unified data access, enhancing data clarity and user experience. Leveraged Django, JavaScript, and Python to streamline bulk data handling, API integration, and complex data presentations.

- **Bulk Data Crawling:** Developed multi-checkbox functionality for bulk contact crawling, integrating Wiza API for large batch processing (up to 2,500 contacts) with optimized rate handling and logging.
- **Apollo API Integration:** Enhanced contact enrichment using Apollo API, enabling concurrent processing with advanced retry logic and data integration.
- **Property Data Page Redesign:** Improved user experience with a card-based layout for search results, refining data display to show the most relevant property per postcode.

Technologies: Python, Django, JavaScript, HTML, RESTful APIs (Wiza, Apollo), Google Cloud Platform, MySQL, PostgreSQL, multi-threaded processing.

Artificial Intelligence Research Intern

May 2024 – August 2024

Lamina Solutions | Toronto, Ontario CA/ Remote

Project Summary: Focused on optimizing a Chatbot testing framework, automating data ingestion, and establishing a scalable GCP-based environment for efficient LLM evaluation.

- **Automated Testing:** Developed Selenium scripts for GUI testing and implemented Query Decomposition to improve Chatbot handling of mixed-context queries.
- **Data & Knowledge Base Integration:** Organized and loaded medical data from 100 PDFs for Chatbot knowledge base; automated creation of 100 test cases, including sentiment and response quality metrics.
- **Cloud Deployment:** Set up a GCP-based test environment for continuous testing, enabling smooth cloud deployments with Docker for Java Spring Boot and Flask.

Technologies: Python, Selenium, Docker, Google Cloud Platform, Java Spring Boot, Flask, advanced performance metrics (ROUGE, BLEU, Sentiment Analysis).

Full Stack Software Developer (Contract)

September 2023 – January 2024

Sherpa | U.S./ Remote

Project Summary: Developed scalable authentication, subscription, and deployment solutions, enhancing user experience on saleslights.com with efficient data handling, API integration, and cloud hosting.

- **User Authentication:** Implemented ETL processes in authentication, including validation and data storage for Login and Signup functionalities, using Node.js and MongoDB.
- **Stripe Integration:** Integrated Stripe API to handle subscription payments (monthly and yearly) for seamless checkout on saleslights.com.
- **UI/UX Implementation:** Translated Figma designs for Login, Signup, and Pricing pages into a responsive interface using ReactJS, improving user engagement.

- **Cross-Platform Testing:** Conducted compatibility tests for a consistent experience across devices (desktop, tablet, mobile).
- **AWS Hosting & Deployment:** Managed domain and sub-domain hosting on AWS, utilizing Route 53, S3, CloudFront for main site, and EC2 for backend.
- **Version Control:** Employed Git and Bitbucket for version control, ensuring smooth team collaboration.

Technologies: Node.js, MongoDB, Stripe API, ReactJS, Figma, AWS (Route 53, S3, CloudFront, EC2), Git, Bitbucket.

Custom Software Engineering Analyst
Accenture Inc. | Mandaluyong City, Philippines

May 2019 – July 2023

Project Summary: Enhanced legacy applications, implemented CI/CD, and maintained code quality, contributing to agile delivery for critical software systems.

- **Legacy Application Enhancement:** Updated Angular 5-based applications with a secure login gateway and dashboard improvements, ensuring robust data access and report functionality.
- **Domain Migration:** Supported application migration to new domains, updating environment endpoints and enabling CORS for improved API functionality.
- **Issue Resolution:** Provided frontend support for JavaScript and Java applications, ensuring minimal downtime and user satisfaction.
- **CI/CD Automation:** Utilized Jenkins to automate build and deployment across Dev, Test, and Prod, streamlining the release process.
- **Angular Development:** Developed an Angular 8-based mobile Insurance Agent app, implementing user stories across sprints, including updates to User Agreement and agent creation pages.
- **Testing & Code Quality:** Developed unit tests in TypeScript, achieving 75% code coverage in Sonar, and reviewed pull requests for quality assurance.
- **Configuration & Scripting:** Created and updated TeamCity configurations using Bash and Python, expediting build activities.
- **Release Management:** Executed ETL for Git commit extraction, integrating package configuration in TeamCity, enhancing release management efficiency.
- **Git Administration:** Managed repository access and monitored sizes, supporting effective project collaboration.
- **Documentation & Coordination:** Maintained a Confluence page for project updates, ensuring clear communication and streamlined project tracking in Agile.

Technologies: Angular (5, 8), JavaScript, Java, Jenkins, TeamCity, Git, Bitbucket, JIRA, Confluence, Python, Bash

Achievements and Recognitions:

- **FY22 Q1 Project Gantimpala Tower Winner**
For integration of the express release notes to the package creation which resulted to 15 minutes savings for each release notes creation.
- **DB Astro Gantimpala Award**
For creating automation task that minimized the space consumption in the SharePoint site and allowed the stakeholders to view the new commits that were part of a certain Full/Delta/Hotfix deployment using log viewer
- **DB Astro LIKHA Awardee – Q3**
For creating automation which resulted to 1 hour and 21 minutes of savings for a task that runs for 2 hours using a step- by-step process to 21 minutes for the Files for Confirmation and 18 minutes for Production Release Notes.

Software Developer 3 as Software Engineer
Pointwest Innovations Corporation | Quezon City, Philippines

March 2015 – September 2015

Project Summary: Led backend Python development and front-end Angular enhancements, implementing automated testing and adding key features to improve software reliability and user experience.

- **Python Development & Code Reviews:** Led Python-focused development, conducting detailed code reviews on Flask-based components.
- **Static Code Testing:** Automated static testing using flake8 in Bamboo with XUnit reporting, ensuring Python code quality and consistent green/OK statuses in CI.
- **Frontend Development in Angular:** Implemented Ratings and Comments functionality, delivering new user stories and UI enhancements for improved interactivity.

Technologies: Python, Flask, Angular, flake8, Bamboo, XUnit.

Associate Software Developer
Canon Information Technologies Inc. | Quezon City, Philippines

July 2011 – August 2014

Project Summary: Contributed to the design and implementation of essential tools, performed ETL operations, and maintained high code quality standards, fulfilling both technical and leadership roles.

- **Full-Stack Tool Development:** Developed four interconnected tools with HTML, CSS, JavaScript (Vanilla) for the frontend and Python for backend, delivering critical releases across projects.
- **Data Processing (ETL):** Executed ETL for a JavaScript application, transforming data between JSON formats using Python for streamlined data handling.
- **Unit Testing & Code Quality:** Created PyUnit test cases, performed static code testing, and analyzed code coverage, ensuring reliable Python-based applications.
- **Technical Leadership:** Managed customer communications, developed test plans, conducted project analysis, assigned tasks, and tracked team progress to ensure successful project delivery.

Technologies: Python, JavaScript (Vanilla), HTML, CSS, JSON, PyUnit